

ABUZER GÜNDÜZ, Ph.D.

Assistant Professor of Mathematics Sakarya University, Mathematics Department Esentepe Campus 54050, Room 302, Serdivan / Sakarya, TURKEY **Email:** abuzergunduz@sakarya.edu.tr **Phone:** +90 535 824 38 66

SUMMARY & RESEARCH INTERESTS

Research Areas: Algebraic Geometry, Commutative Algebra, Singularity Theory, Quiver Representations. Experienced mathematician and researcher with a solid background in algebraic geometry and singularity theory. Recently completed a visiting scholarship at the University of Illinois Chicago (UIC) focusing on advanced topics in algebraic geometry. Highly motivated to bridge classical singularity theory with modern gauge-theoretic techniques and quiver representations.

EDUCATION

- Ph.D. in Mathematics (Algebraic Geometry)** | Dec 2021 Yildiz Technical University, Istanbul, Turkey
 - Thesis:** Detection methods of asymptotic critical values of polynomial mappings
 - Supervisors:** Prof. Dr. Susumu Tanabe
- M.Sc. in Mathematics** | 2014 Marmara University, Istanbul, Turkey
 - Thesis:** Linear Free divisor and quiver representations
 - Supervisors:** Prof. Dr. Meral Tosun
- B.Sc. in Mathematics** | 2006 Istanbul University, Istanbul, Turkey
- High School Diploma** | 2000 Diyarbakır Science High School, Turkey (*Admitted among the top 100 students nationwide*)

ACADEMIC & RESEARCH EXPERIENCE

- Assistant Professor of Mathematics** | 2020 – Present Sakarya University, Mathematics Department, Turkey
- Visiting Scholar (Postdoctoral Researcher)** | Jul 2023 – Jul 2024 University of Illinois Chicago (UIC), Mathematics Department, IL, USA
 - Supervised by Prof. Dr. Izzet Coskun; funded by TÜBİTAK 2219 Fellowship.
- Research Assistant** | 2020 – 2022 Sakarya University, Mathematics Department, Turkey
- Lecturer** | 2017 – 2020 Istanbul Medipol University, Turkey

GRANTS & FELLOWSHIPS

- TÜBİTAK Grant 2219 (Postdoctoral Fellowship):** "Grassmann Varieties and Singularity Theory" (Jul 2023 - Jul 2024).
- TÜBİTAK Grant 1001 (Research Project):** "Period Integrals for Algebraic Varieties" (Mar 2017 - Mar 2020).
- TÜBİTAK Grant 2209:** "r-ideals in Amalgamated ring" (Ongoing).

PUBLICATIONS

Books

- **Gündüz, A.** (2021). *Ring Theory (Halka Teorisi)*. Nobel Akademik Publication, Ankara, Turkey.

Journal Articles & Preprints

- Gunduz, A. (2025). On 2-absorbing submodules and their Amalgamations. *arXiv preprint arXiv:2510.09863*.
- Gündüz, A., Naji, O. A., & Özen, M. (2025). Sr-ideals in Commutative Rings. *arXiv preprint arXiv:2505.19198*.
- Coskun, I., & Gündüz, A. (2024). PGL orbits on products of flag varieties. *arXiv preprint arXiv:2412.04870*.
- Gündüz, A. (2024). E-exact sequence and some results. *İstanbul Ticaret Üniversitesi Fen Bilimleri Dergisi*, 23(46), 319-328.
- Gündüz, A. (2023). On 1-absorbing Prime and 1-absorbing Primary Ideals of Amalgamations. *Proceedings of the Bulgarian Academy of Sciences*, 76(9), 1317-1325.
- Naji, O. A., Gündüz, A., Özen, M., & Tekir, Ü. (2023). On uniformly 1-absorbing primary ideals. *Publ. Math. Debrecen*, 102(1-2), 81-93.
- Tanabé, S., & Gunduz, A. (2023). On curves approaching asymptotic critical value set of a polynomial map. *Kumamoto Journal of Mathematics*, 36, 17-40.
- Gündüz, A., & Naji, O. (2022). A Note On E-Injective Modules. *Sakarya University Journal of Science*, 26(6), 1262-1266.
- Tanabé, S., Gündüz, A., & Ersoy, B. A. (2021). On real curve construction for asymptotic critical value set of a polynomial map. *Comptes rendus de l'Académie bulgare des Sciences*, 74(5).

CONFERENCES & PRESENTATIONS

- **Gündüz, A.** (2024). *1-absorbing prime and 1-absorbing primary ideals of Amalgamation*. ICOMAA 2024.
- **Gündüz, A.** (2024). *2-absorbing primary ideal of Amalgamations*. International Thrace Scientific Research Congress.
- **Gündüz, A.** (2023). *2-absorbing prime ideal of Amalgamations*. ICAR 2025 (Oral Presentation).
- **Gündüz, A.** (2021). *E-Exact Sequence and Some Results*. ICOMAA 2021.
- **Gündüz, A.** (2021). *U-Exact Sequence and Some Results*. ICOMAA 2021.
- **Gündüz, A.** (2020). *On the real curve method for asymptotic critical value set of polynomial map*. ICOLES 2020.
- **Gündüz, A.** (2020). *On Uniformly 1-Absorbing Primary Ideals*. ICMRS 2020.
- **Gündüz, A.** (2019). *Detection of Asymtotic Critical Values of Polynomial Mapping*. ICOMAA 2019.

ACADEMIC SERVICES & ORGANIZATION

- **Organizing Committee Member:** ICOMAA (International Conference on Mathematics and Applications) 2020, 2021, 2022, 2023, 2024, 2025.
 - **Organizer:** Singularities of polynomial mappings and monodromy Summer School.
<https://matematik.sakarya.edu.tr/tr/duyuru/goster/148957/singularities-of-polynomial-mappings-and-monodromy-summer-school>
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REFERENCES

- Prof. Dr. Susumu Tanabe Ph.D. Thesis Supervisor, Moscow Institute of Physics and Technology (MIPT), Dept. of Discrete Mathematics **Email:** [tanabe.s@mipt.ru]
- Prof. Dr. Izzet Coskun *Postdoctoral Research Supervisor* Department of Mathematics, Statistics, and Computer Science University of Illinois Chicago (UIC), Chicago, IL, USA **Email:** [icoskun@uic.edu]